Inflammatory Bowel Disease 29 (IBD29): A Comprehensive Disease Characteristics Report

Disease: Inflammatory Bowel Disease 29 (IBD29) **MONDO ID:** MONDO:0054849 · **OMIM:** 618077 · **Causal gene: INAVA** (Innate Immunity Activator; formerly *C1orf106*; HGNC:25599; UniProt Q3KP66; Entrez 55765; Ensembl ENSG00000163362; locus 1q32.1)

Summary

Inflammatory Bowel Disease 29 (IBD29; OMIM 618077; MONDO:0054849) is a multifactorial, polygenic susceptibility form of inflammatory bowel disease caused by variation in the gene INAVA (Innate Immunity Activator, formerly named *C1orf106*). It is not a distinct clinical syndrome with a unique presentation; rather, it is a molecularly defined genetic contribution to common IBD, so affected individuals present clinically as either **Crohn disease (CD)** or **ulcerative colitis (UC)**. UniProt curates the entry as disease variant DI-05306: "susceptibility associated with variants affecting the gene." Two variants anchor the genetic architecture: a common intronic regulatory SNP **rs7554511** (risk allele C) that lowers INAVA transcription and expression, and a rare coding missense variant **p.Tyr333Phe (rs41313912)** that destabilizes the protein.

INAVA encodes a **dual-function cytoplasmic/nuclear scaffold protein** that sits at the intersection of two processes central to IBD pathogenesis: the intestinal epithelial barrier and innate immune signaling. In **epithelial cells**, INAVA stabilizes adherens and tight junctions by restraining cytohesin-1/ARNO–driven ARF6 activation and by organizing cortical actin through a ROCK-dependent mechanism; loss of INAVA produces a "leaky gut" with increased permeability and heightened sensitivity to TNF-α and enteric pathogens. In **myeloid cells (macrophages)**, INAVA is required for optimal pattern-recognition-receptor (PRR)–induced MAPK and NF-κB activation, cytokine secretion, and intracellular bacterial clearance, acting through recruitment of 14-3-3τ and amplification of TRAF6-dependent polyubiquitination

within biomolecular condensates. The IBD-risk genotypes reduce both functions, so that ordinary microbial exposure in a genetically susceptible gut is converted into **chronic, relapsing mucosal inflammation**.

There is **no INAVA-specific therapy**; IBD29 is managed with standard IBD pharmacotherapy (5-aminosalicylates, corticosteroids, immunomodulators, and biologics such as anti-TNF agents). A pharmacogenomic signal links *C1orf106* polymorphisms to variability in infliximab trough levels, and the underlying cell biology nominates druggable nodes (p38α, mTOR, and Rho-ROCK pathways) that shift INAVA between its inflammatory-condensate and barrier-promoting functions. This report synthesizes seven confirmed findings from five iterations of investigation, grounded in six primary mechanistic/clinical papers.

---

1. Disease Information

Overview. IBD29 is a Mendelian-registry-labeled but mechanistically **polygenic/multifactorial susceptibility** form of inflammatory bowel disease. The number "29" is a serial OMIM susceptibility-locus designation and does **not** indicate a distinct clinical phenotype; the "IBD##" series catalogs genes and loci that confer susceptibility to CD and/or UC. For IBD29 the responsible gene is **INAVA / C1orf106**. Affected individuals present with the standard spectrum of chronic relapsing intestinal inflammation (CD or UC).

Key identifiers.

Resource	Identifier
OMIM (phenotype)	618077
MONDO	MONDO:0054849
Causal gene	INAVA (HGNC:25599; Entrez 55765; Ensembl ENSG00000163362)
UniProt (protein)	Q3KP66 (663 aa)
UniProt disease	DI-05306
Cytogenetic locus	1q32.1
MeSH (parent)	Inflammatory Bowel Diseases (D015212)
ICD-10	K50 (Crohn disease) / K51 (ulcerative colitis) — IBD29 inherits the code of the manifesting subtype
ICD-11	DD70 (Crohn disease) / DD71 (ulcerative colitis)

Synonyms / alternative names. The causal gene has been known as *C1orf106* (chromosome 1 open reading frame 106) and *INAVA*. No disease-specific synonyms are recorded for IBD29 beyond "inflammatory bowel disease 29" and the OMIM designation; the clinical presentations are referred to by their subtype names (Crohn disease, ulcerative colitis).

Source of information. The information is derived from **aggregated, disease-level and gene-level resources** (OMIM, UniProt, HGNC, GWAS meta-analyses) and from **mechanistic studies in human macrophages, human intestinal epithelial cells, patient-derived hiPSC intestinal spheroids, and knockout mice**, rather than from a single patient EHR cohort.

MONDO term suggestion: MONDO:0054849 (inflammatory bowel disease 29); parent MONDO:0005265 (inflammatory bowel disease).

2. Etiology

Primary causes. IBD29 is **genetically driven susceptibility acting in concert with the gut microbial environment**. It is not infectious or toxin-caused in a classical sense; rather, inherited reduction of INAVA function lowers two protective barriers — epithelial integrity and myeloid antimicrobial signaling — so that the normal luminal microbiota triggers and sustains inflammation (gene–environment interaction).

Genetic risk factors. - **rs7554511** — common intronic tag SNP at 1q32.1 (GRCh38 chr1:200,908,434 C>A). The **risk allele is C (the common allele)**; the minor allele A has MAF ≈ 0.21 in European gnomAD/TOPMED and ≈ 0.14 globally. Risk carriers (rs7554511-C) express **lower INAVA RNA and protein**, attributable to decreased transcription driven by the intronic region containing the C variant (PMID: 28436939). - **p.Tyr333Phe (rs41313912)** — rare coding missense variant (NC_000001.11:g.200908897A>G/T), UniProt feature VAR_080249, annotated "in IBD29; decreases protein stability." Minor-allele (T) frequency ≈ 0.4% in 1000 Genomes, ≈ 1% in ALFA. - The INAVA/C1orf106 locus was identified in genome-wide association meta-analyses of IBD and sits within the broader polygenic IBD risk architecture (>240 IBD loci known).

Environmental risk factors (general IBD, applicable to IBD29). Enteric microbial exposure/dysbiosis, smoking (risk factor for CD, protective for UC), diet (Western, low-fiber), NSAID use, enteric infections, and early-life antibiotic exposure. Food insecurity is associated with higher odds of IBD (adjusted OR 1.78, 95% CI 1.42–2.23; PMID: 42847277). Family history and sex are standard demographic modifiers of IBD risk.

Protective factors. The ancestral/minor allele of rs7554511 (A) is associated with higher INAVA expression and thus lower risk. No INAVA-specific environmental protective factor is established. In general IBD, breastfeeding, high dietary fiber, and (for UC) smoking and appendectomy are reported protective associations; the evidence for a protective role of *Helicobacter pylori* in IBD remains inconsistent (PMID: 42895565).

Gene–environment interaction. The mechanistic core of IBD29 is a GxE interaction: reduced INAVA function is clinically silent until the epithelium is challenged. C1orf106-deficient mice are **hypersensitive to TNF-α–induced increases in epithelial permeability** and show increased susceptibility to intestinal pathogens — i.e., the phenotype emerges on microbial/cytokine challenge (PMID: 29420262; PMID: 31022698).

3. Phenotypes

Because IBD29 manifests as CD or UC, its phenotype spectrum is that of IBD. Core and extraintestinal phenotypes with suggested HPO terms:

Phenotype	Type	HPO term	Typical characteristics
Chronic diarrhea	Symptom	HP:0002028	Childhood–adult onset; episodic/ relapsing; very common
Hematochezia / bloody stool	Sign	HP:0002573	Prominent in UC; relapsing
Abdominal pain	Symptom	HP:0002027	Common; fluctuating
Weight loss	Sign	HP:0001824	Moderate–severe in active CD
Malabsorption	Lab/clinical	HP:0002024	More in CD (small-bowel)
Intestinal/perianal fistula	Physical manifestation	HP:0100275	CD complication (penetrating, B3)
Intestinal obstruction	Physical manifestation	HP:0005214	CD stricturing (B2)
Elevated CRP	Lab abnormality	HP:0011227	Tracks disease activity
Elevated ESR	Lab abnormality	HP:0003565	Tracks disease activity
Anemia	Lab abnormality	HP:0001903	Iron-deficiency / chronic disease
Arthritis	Extraintestinal	HP:0001369	Peripheral/axial
Uveitis	Extraintestinal	HP:0000554	Ocular
Erythema nodosum	Extraintestinal	HP:0012219	Cutaneous
Primary sclerosing cholangitis	Extraintestinal	HP:0030151	Hepatobiliary (esp. UC)
Increased colorectal cancer risk	Complication	HP:0003003	Long-standing colitis

Onset / severity / progression / frequency. Onset is most often in adolescence/young adulthood with a second smaller peak later in life; severity is **variable**; the course is characteristically **relapsing-remitting (episodic)**. Frequencies above are qualitative for IBD at large (no IBD29-genotype-stratified phenotype frequencies are published).

Quality-of-life impact. IBD substantially reduces health-related quality of life; developmental, emotional, and cognitive factors modulate subjective symptom burden (PMID: 42652338). Fatigue, urgency, diet restriction, and psychosocial burden are major contributors.

---

4. Genetic / Molecular Information

Causal gene. INAVA (C1orf106), 1q32.1; Entrez 55765; HGNC:25599; UniProt Q3KP66 (663 aa); Ensembl ENSG00000163362.

Protein features. INAVA contains a **coiled-coil region (aa 118–147)**, **three nuclear localization signals**, large **intrinsically disordered regions**, and a bifunctional **DUF3338 domain renamed CUPID**. It localizes to both cytoplasm and nucleus and translocates to the nucleus upon NOD2 stimulation.

Pathogenic / risk variants.

Variant	rsID	Type	Classification	Frequency	Functional consequence
Intronic tag SNP (risk allele C)	rs7554511	Regulatory (intronic)	Common risk allele	minor allele A MAF ~0.21 EUR, ~0.14 global	↓ transcription → ↓ INAVA RNA/ protein (loss-of-function-leaning)
p.Tyr333Phe	rs41313912	Missense (coding)	"in IBD29" (UniProt VAR_080249)	minor allele T ~0.4% 1000G, ~1% ALFA	↓ protein stability; ↑ ubiquitination/ turnover (loss of function)

- **Somatic vs germline:** germline susceptibility variants (not somatic).
- **Functional consequence:** both variants converge on **reduced INAVA function/dosage** — rs7554511-C lowers expression; p.Y333F destabilizes the protein. The IBD-risk coding

variant **increases C1orf106 ubiquitination and turnover with consequent functional impairments** (PMID: 29420262).

Modifier genes. No INAVA-specific modifier genes are established; IBD29 operates within the broader polygenic IBD background (NOD2, IL23R, ATG16L1, IL10, etc.).

Epigenetic information. No IBD29/INAVA-specific DNA-methylation or histone-modification data are established; the rs7554511 effect is transcriptional/regulatory.

Chromosomal abnormalities. None described for IBD29; it is a single-gene susceptibility locus, not a structural/aneuploidy disorder.

Gene/GO annotation suggestions: INAVA (HGNC:25599); GO:0032088 (negative regulation of NF-κB), GO:0038061 (NIK/NF-κB signaling), GO:0007010 (cytoskeleton organization), GO:0045216 (cell-cell junction organization).

---

5. Environmental Information

- **Environmental factors.** The dominant environmental input is the **enteric microbiota**; INAVA loss increases susceptibility to intestinal pathogens and impairs intracellular bacterial clearance in macrophages. No specific toxin/radiation/occupational exposure is implicated for IBD29 beyond general IBD risk factors.
- **Lifestyle factors.** General IBD lifestyle associations apply: smoking (divergent for CD vs UC), Western diet, NSAIDs, stress; food insecurity associated with higher IBD odds (PMID: 42847277).
- **Infectious agents.** No single pathogen causes IBD29. The commensal/pathobiont microbiota acts as the environmental trigger converting genetic susceptibility to inflammation; enteric pathogens precipitate barrier breakdown in INAVA-deficient models.

CHEBI suggestion: tumor necrosis factor (challenge agent in models); lipopolysaccharide (PRR ligand); muramyl dipeptide (NOD2 ligand).

---

6. Mechanism / Pathophysiology

Ordered causal chain

1. **Inheritance of an INAVA risk genotype** (rs7554511-C lowering expression, and/or p.Y333F destabilizing the protein) → **leads to** reduced functional INAVA protein dosage in intestinal epithelial and myeloid cells. *(Demonstrated: expression and stability data, PMIDs 28436939, 29420262.)*

2. **Branch A — Epithelial barrier arm.** Reduced INAVA → **fails to restrain** cytohesin-1–dependent ARF6 activation (INAVA normally promotes cytohesin-1 degradation) and **impairs** CUPID–ARNO-driven lateral-membrane F-actin assembly and ROCK-dependent cortical actin-belt dynamics → **results in** destabilized adherens/tight junctions, defective cell polarity and migration → **leads to** increased intestinal epithelial permeability ("leaky gut"). *(Demonstrated: PMIDs 29420262, 30355448, 40161582.)*

3. **Branch B — Myeloid innate-immune arm.** Reduced INAVA → **fails to recruit** 14-3-3τ and **fails to amplify** TRAF6-dependent polyubiquitination within IL-1β-induced biomolecular condensates → **results in** sub-optimal MAPK and NF-κB activation, reduced cytokine secretion, and **impaired intracellular bacterial clearance**. *(Demonstrated: PMIDs 28436939, 30355448, 34251416.)*

4. **Convergence under microbial/cytokine challenge.** Leaky epithelium (Branch A) **permits** increased luminal-microbe translocation; impaired myeloid clearance (Branch B) **fails to eliminate** translocated microbes → **leads to** persistent antigenic/microbial stimulation of the lamina propria. *(Inferred synthesis of both arms, supported by TNF-α hypersensitivity in vivo,*

   31022698

   .)

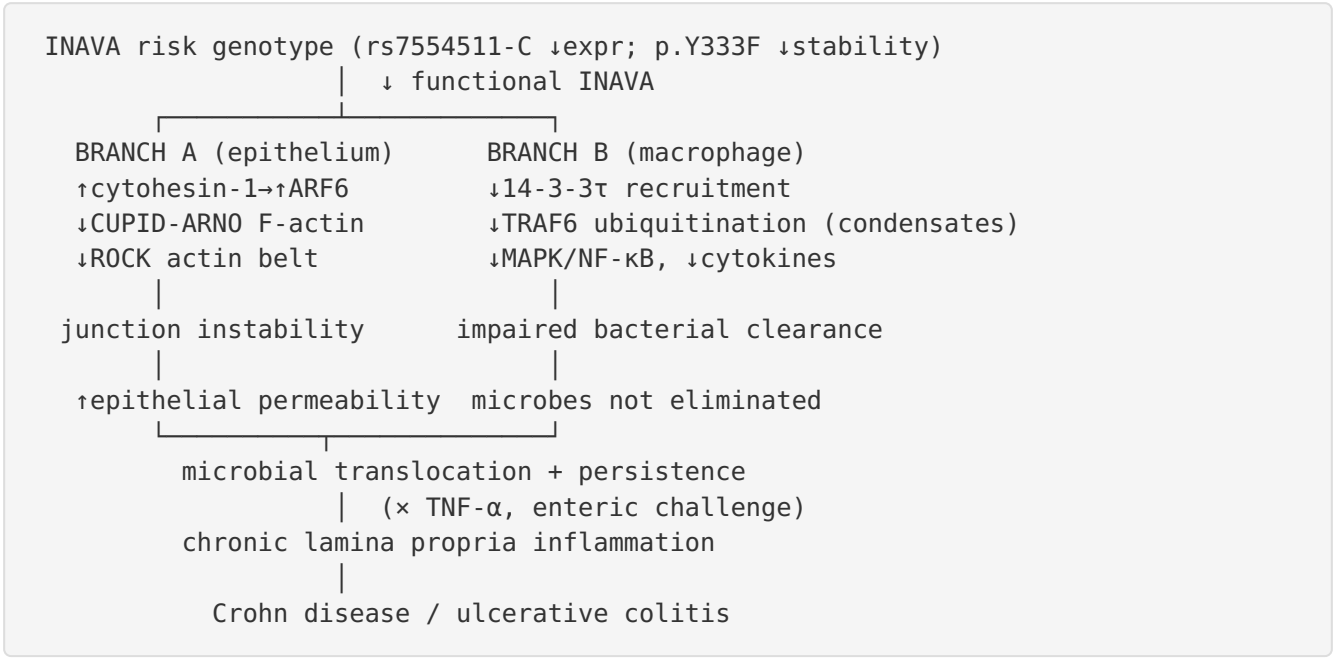
5. **Persistent stimulation → leads to** chronic mucosal immune activation, cytokine-driven tissue injury, and the relapsing-remitting clinical phenotype of CD/UC. *(Inferred from model phenotypes + IBD pathophysiology.)*

Mechanistic detail by category

- **Molecular pathways.** Cytohesin-1/ARNO → **ARF6** GTPase signaling (junctional actin); **Rho–ROCK** (cortical actin belt); **MAPK** and **NF-κB** (PRR-driven inflammation); **TRAF6** polyubiquitination; **p38α/MAPK** and **mTOR** (condensate dynamics).

- **Cellular processes.** Cell–cell junction assembly (GO:0034329), actin cytoskeleton organization (GO:0030036), cell polarity/migration, inflammatory signaling, intracellular bacterial killing, biomolecular-condensate (liquid–liquid phase separation) assembly and resolution.
- **Protein dysfunction.** p.Y333F is a **destabilizing loss-of-function** variant (increased ubiquitination/turnover); rs7554511-C causes **reduced dosage**. INAVA is a bifunctional scaffold; the CUPID/DUF3338 domain mediates both the barrier (ARNO binding) and inflammatory (TRAF6 amplification) activities.
- **Immune system involvement.** Defective innate antimicrobial response in macrophages (PRR→MAPK/NF-κB, bacterial clearance) rather than classical autoimmunity; downstream chronic inflammation.
- **Tissue-damage mechanisms.** Barrier failure → microbial translocation → sustained inflammation → mucosal injury; TNF-α amplifies permeability.
- **Molecular profiling.** Risk-allele macrophages show lower INAVA RNA/protein and reduced PRR-induced cytokines (PMID: 28436939); INAVA condensates contain ubiquitin and E3 ligase βTrCP2 (PMID: 34251416).

Causal-chain diagram



GO/CL term suggestions: GO:0043123 (positive regulation of NF-κB), GO:0007015 (actin filament organization), GO:0034329 (cell junction assembly), GO:0071345 (cellular response to cytokine stimulus); CL:0000235 (macrophage), CL:0002253 (epithelial cell of large intestine), CL:0000160 (goblet cell).

---

7. Anatomical Structures Affected

- **Organ level.** Primary: **intestine** (colon and/or small bowel) — UBERON:0000160 (intestine), UBERON:0001155 (colon), UBERON:0002108 (small intestine). Secondary/complication: perianal region (fistulae), hepatobiliary tract (PSC), joints, eyes, skin. Body system: **digestive system** (UBERON:0001007), with immune-system involvement.
- **Tissue / cell level.** Intestinal **epithelium** (UBERON:0001277 intestinal mucosa; UBERON:0000483 epithelium) and **lamina propria** immune infiltrate. Key cells: **intestinal epithelial cells** (CL:0002253), **macrophages** (CL:0000235), and junction-forming enterocytes.
- **Subcellular level.** INAVA localizes to **cytoplasm** and **nucleus** (GO:0005737, GO:0005634); functions at the **lateral plasma membrane / cell–cell junction** (GO:0016323, GO:0005912 adherens junction) and in **cytosolic biomolecular condensates** (GO:0036464).
- **Localization / lateralization.** Mucosal inflammation is typically **patchy/segmental in CD** and **continuous, diffuse colonic in UC**; not a lateralized disease.

---

8. Temporal Development

- **Onset.** Chronic/insidious; most commonly adolescence to young adulthood, consistent with IBD. No IBD29-specific age-of-onset data beyond general IBD.
- **Progression.** CD can progress from inflammatory (B1) to stricturing (B2) or penetrating (B3) behavior; UC can extend proximally. Progression rate is **variable**.
- **Course pattern.** Characteristically **relapsing-remitting (episodic)**; chronic lifelong.
- **Remission.** Both spontaneous and, more reliably, **treatment-induced** remission occur.
- **Critical periods / intervention windows.** Early effective control of inflammation ("treat-to-target") and barrier protection are the principal opportunities; mechanistically, challenge-dependent barrier failure suggests windows around enteric infection/antibiotic exposure.

9. Inheritance and Population

- **Epidemiology.** IBD29 has no separate prevalence/incidence; it contributes to the IBD population. IBD affects >**7 million people worldwide**; incidence has plateaued in high-income countries but is rising in lower- and middle-income regions (PMID: 42953891).
- **Inheritance pattern. Multifactorial / polygenic susceptibility** (not simple Mendelian), despite the OMIM "IBD29" designation. INAVA is one susceptibility locus among many.
- **Penetrance / expressivity. Incomplete penetrance** and **variable expressivity** — the risk genotype is common but most carriers do not develop IBD; manifestation requires additional genetic and environmental hits.
- **Anticipation / mosaicism / founder effects.** Not applicable/not described.
- **Consanguinity.** Not a driver (common-variant susceptibility, not recessive rare disease).
- **Allele frequency.** rs7554511 risk allele C is the **common** allele (minor allele A MAF ~0.14–0.21); p.Y333F is rare (~0.4–1%).
- **Population demographics.** The locus was defined in predominantly European GWAS meta-analyses; the INAVA region shows pleiotropy with extraintestinal immune traits, with chromosome-1 enrichment in IBD–iridocyclitis genetic correlation (PMID: 39472800). Sex ratio and geographic distribution follow general IBD patterns.

10. Diagnostics

IBD29 is diagnosed as its manifesting IBD subtype; genetic findings are supportive/research-level, not stand-alone diagnostic.

- **Laboratory tests.** CRP (HP:0011227), ESR (HP:0003565), CBC (anemia HP:0001903), **fecal calprotectin/lactoferrin** (mucosal inflammation markers), albumin, iron studies.
- **Biomarkers.** Fecal calprotectin; serologies (ASCA, pANCA) for CD/UC distinction; genotype (rs7554511, rs41313912) as research biomarkers.
- **Imaging.** CT/MR enterography, intestinal ultrasound.
- **Endoscopy / biopsy / pathology.** Ileocolonoscopy with biopsy is the diagnostic cornerstone; histology shows chronic active inflammation, crypt architectural distortion, and (CD) non-caseating granulomas.

- **Genetic testing.** WES/WGS or IBD gene panels can detect INAVA variants; single-gene testing (rs7554511, p.Y333F) has research/pharmacogenomic value but is **not standard clinical diagnostic** for common IBD. CMA, karyotyping, FISH, mtDNA, and repeat-expansion testing are **not applicable**.
- **Omics-based diagnostics.** Transcriptomic risk-allele effects on INAVA expression demonstrated in macrophages; not a routine clinical assay.
- **Clinical criteria / differential diagnosis.** ECCO/ACG guidelines; differentials include infectious colitis, ischemic colitis, microscopic colitis, Behçet disease, intestinal TB, and drug-induced enteropathy.
- **Screening.** No population genetic screening for IBD29; colorectal cancer surveillance for long-standing colitis.

---

11. Outcome / Prognosis

- **Survival / mortality.** IBD (and thus IBD29) is generally **not life-limiting** with modern therapy; mortality is driven by complications (surgery, severe flares, colorectal cancer, thromboembolism). No IBD29-specific survival figures.
- **Morbidity / function / QoL.** Substantial — fatigue, urgency, surgery, disability, and reduced quality of life; psychosocial and developmental factors modulate symptom burden (PMID: 42652338).
- **Complications.** Strictures, fistulae, abscesses, obstruction, perforation, malnutrition, anemia, extraintestinal manifestations, and increased colorectal cancer risk (HP:0003003).
- **Recovery potential.** Remission achievable with therapy; disease is chronic with relapse risk.
- **Prognostic factors.** Disease extent/behavior, age at onset, smoking, early biologic response, and biomarker trajectories (CRP, calprotectin). *C1orf106* genotype is a candidate pharmacogenomic predictor of infliximab exposure (see §12).

---

12. Treatment

No INAVA-targeted therapy exists; IBD29 is treated with standard IBD therapy matched to subtype and severity.

Modality	Examples	NCIT suggestion
Aminosalicylates	mesalamine (5-ASA)	NCIT:C29281 (aminosalicylic acid class)
Corticosteroids	prednisone, budesonide	NCIT:C442 (corticosteroid)
Immunomodulators	azathioprine, 6-MP, methotrexate	NCIT:C569 (immunosuppressant)
Anti-TNF biologics	infliximab, adalimumab	NCIT:C1965 (infliximab)
Anti-integrin	vedolizumab	NCIT:C90555
Anti-IL-12/23	ustekinumab	NCIT:C82407
JAK inhibitors	tofacitinib, upadacitinib	NCIT:C116834
Surgery	resection, colectomy, fistula repair	NCIT:C15329 (surgery)

- **Pharmacogenomics.** Polymorphisms in *C1orf106* (together with *IL1RN* and *IL10*) contribute to variability in post-induction **infliximab trough levels** in a cohort of 189 Crohn disease patients (multivariate model AUROC 0.758 training / 0.733 testing) (PMID: 33163192).
- **Mechanistically nominated druggable nodes. p38α and mTOR inhibitors** enhance resolution of INAVA inflammatory condensates; **Rho-ROCK inhibitors** recruit INAVA to intercellular junctions (pro-barrier), and ROCK inhibition rescues C1ORF106-knockdown actin/polarity defects (PMID: 34251416; PMID: 40161582). These are **experimental/preclinical** leads, not approved IBD29 therapies.
- **Advanced/RNA/gene/cell therapy.** No INAVA-specific gene, cell, or RNA therapy is in clinical use.
- **Supportive care.** Nutrition, iron repletion, smoking cessation (CD), vaccination before immunosuppression.
- **Strategy.** Treat-to-target with step-up or top-down algorithms; genotype-guided dosing remains investigational.

13. Prevention

- **Primary prevention.** No specific primary prevention for IBD29; general IBD risk-factor modification (smoking cessation for CD, dietary measures, judicious antibiotic/NSAID use).
- **Secondary prevention.** Early detection of flares via fecal calprotectin monitoring; colorectal cancer surveillance colonoscopy in long-standing colitis.
- **Tertiary prevention.** Maintenance therapy to prevent relapse, strictures, and surgery; bone, thrombosis, and infection prophylaxis during immunosuppression.
- **Immunization.** Age-appropriate and pre-biologic vaccination (not disease-causing but part of IBD management).
- **Genetic screening / counseling.** No carrier or prenatal screening is indicated for a common polygenic susceptibility variant; genetic counseling emphasizes **multifactorial, low-penetrance** risk.
- **Public-health / environmental.** General IBD measures (diet, microbiome health). No specific environmental intervention established for IBD29.

14. Other Species / Natural Disease

- **Taxonomy / orthologs.** INAVA orthologs: mouse *Inava* (Entrez 67313; MGI:1921579; chr1 E4), rat *Inava* (289399), chimpanzee (101058404), chicken (430729), and zebrafish paralogs (563192, 567844). Human gene ENSG00000163362.
- **Natural disease in other species.** No well-characterized spontaneous INAVA-associated enteropathy is catalogued in companion animals or wildlife (OMIA); IBD-like disease in dogs/cats is multifactorial and not ascribed to INAVA.
- **Comparative biology.** The barrier and innate-immune functions of INAVA appear conserved in mouse (knockout phenotypes mirror human barrier susceptibility), supporting evolutionary conservation of the mechanism.
- **Transmission / zoonosis.** Not applicable — IBD29 is a non-communicable, genetically influenced disease.

15. Model Organisms

Model	Type	Key phenotype	Reference
C1orf106/Inava knockout mouse	Mammalian, germline KO	Intestinal epithelial **barrier defects**, increased susceptibility to intestinal pathogens	PMID: 29420262
C1orf106-deficient mouse	Mammalian KO	**Hypersensitivity to TNF-α–induced epithelial permeability** with increased diarrhea	PMID: 31022698
Human colonic epithelial cell knockdown	In vitro / cellular	Impaired actin belt, cell polarity, barrier	PMID: 40161582
Patient-derived **hiPSC intestinal spheroids** (p.Y333F)	iPSC-derived organoid	Cell-polarity defects reproduced by the IBD variant; rescued by ROCK inhibition	PMID: 40161582
Primary human macrophages	In vitro (human)	PRR/IL-1β signaling, 14-3-3τ recruitment, condensate dynamics	PMID: 28436939; PMID: 30355448

- **Genetic model types available.** Knockout mice; cell-line knockdowns; patient-specific iPSC-derived intestinal models carrying p.Y333F.
- **Phenotype recapitulation.** Strong for the **barrier arm** (mouse KO leaky-gut and pathogen susceptibility; iPSC spheroid polarity defects) and for the **myeloid signaling arm** (human macrophage assays).
- **Limitations.** Mouse KO is a complete-loss model whereas human risk is partial-dosage; spontaneous colitis is challenge-dependent, not fully autonomous; the common human risk allele is regulatory, so knockouts overstate effect size.
- **Resources.** MGI (MGI:1921579), IMPC/IMSR for mouse lines; Cellosaurus for iPSC lines.

Mechanistic Model / Interpretation

INAVA is best understood as a **molecular rheostat that couples epithelial barrier integrity to innate-immune antimicrobial competence**, using a single bifunctional CUPID (DUF3338) domain. In the resting epithelium, CUPID binds the cytohesin ARF-GEF **ARNO** to build lateral-membrane F-actin that underpins cell–cell junctions, and INAVA restrains cytohesin-1/ARF6 to keep adherens junctions stable; ROCK-dependent actin-belt dynamics maintain polarity. Upon inflammatory stimulation (IL-1β, NOD2 agonists), INAVA switches modes — relocating into **biomolecular condensates** where CUPID amplifies TRAF6-dependent polyubiquitination (with βTrCP2 and ubiquitin) to drive MAPK/NF-κB signaling and bacterial clearance. ARNO negatively regulates this inflammatory arm, embodying a built-in barrier-versus-inflammation toggle.

IBD29 genotypes **turn the rheostat down**: rs7554511-C lowers total INAVA, and p.Y333F accelerates its degradation. The two arms fail together — a leakier epithelium admits more microbes while macrophages clear them less efficiently — so that under the everyday challenge of the gut microbiota (and amplifying signals like TNF-α), the system tips into chronic inflammation. This dual-hit, challenge-dependent model explains why the risk allele is common yet low-penetrance, why the phenotype is the familiar CD/UC spectrum rather than a unique syndrome, and why both barrier-directed (ROCK) and inflammation-directed (p38α/mTOR) interventions are mechanistically plausible.

Evidence Base

PMID	Role	Contribution
28436939	Core (human macrophage)	INAVA required for PRR-induced MAPK/NF-κB, cytokines, bacterial clearance via 14-3-3τ; rs7554511-C carriers have lower INAVA RNA/protein via decreased intronic-driven transcription
29420262	Core (Science; mouse + mechanism)	C1orf106 stabilizes adherens junctions by promoting cytohesin-1 degradation → limiting ARF6; KO mice have barrier defects/pathogen susceptibility; IBD risk variant ↑ubiquitination/turnover
31022698	Supporting (mouse)	C1orf106 interacts with cytohesins; KO mice hypersensitive to TNF-α–induced permeability with diarrhea
30355448	Core (eLife; mechanism)	CUPID/DUF3338 domain binds ARNO for barrier F-actin and amplifies TRAF6 polyubiquitination in IL-1β puncta
34251416	Supporting (J Cell Biol)	INAVA puncta are biomolecular condensates with ubiquitin + βTrCP2; p38α/mTOR inhibitors resolve them; ROCK inhibition drives barrier function
40161582	Supporting (iPSC/ cellular)	C1ORF106 KD impairs actin belt/polarity/migration; p.Y333F reproduces defect; ROCK inhibition rescues
33163192	Supporting (pharmacogenomics)	C1orf106 polymorphisms contribute to infliximab trough-level variability (AUROC ~0.73–0.76)
39472800	Context	IBD genetic pleiotropy with iridocyclitis, chromosome-1 enrichment
42953891	Context (epidemiology)	IBD affects >7 million worldwide; rising incidence in LMICs

Key verbatim support. - "Upon PRR stimulation, INAVA was required for optimal MAPK and NF-κB activation, cytokine secretion, and intracellular bacterial clearance. INAVA recruited 14-3-3τ..."
(P 28436939)
- "C1orf106 regulates adherens junction stability by regulating the degradation of cytohesin-1... By limiting cytohesin-1-dependent ARF6 activation, C1orf106 stabilizes adherens junctions."

(P 29420262)

- "Both activities require INAVA's DUF3338 domain (renamed CUPID). CUPID stably binds the cytohesin ARF-GEF ARNO to effect lateral membrane F-actin assembly..."

(P 30355448)

- "IL-1β-induced INAVA puncta are biomolecular condensates... The condensates contain ubiquitin and the E3 ligase βTrCP2."

(P 34251416)

- "an inhibition of ROCK rescues the actin belt and cell polarity phenotypes in C1ORF106 KD cells..."

(P 40161582)

Limitations and Knowledge Gaps

1. **No IBD29-genotype-stratified clinical data** — phenotype frequencies, age of onset, severity, and prognosis are inferred from general IBD, not from INAVA-genotype-defined cohorts.
2. **Dosage vs knockout gap** — most in vivo mechanism comes from complete-loss mouse models, whereas the common human risk is partial expression reduction; effect sizes may be overstated.
3. **Regulatory variant mechanism** — the precise transcription-factor/enhancer basis of the rs7554511 effect is not fully resolved; no methylation/chromatin data specific to IBD29.
4. **Epidemiology is non-specific** — no prevalence/incidence, penetrance estimate, or carrier-adjusted risk is quantified for the INAVA locus alone.
5. **Druggable nodes are preclinical** — p38α/mTOR/ROCK leads derive from cellular/condensate biology; none is validated clinically for IBD29.
6. **Pharmacogenomic finding is single-cohort** (n=189) and needs replication before clinical use.
7. **Limited literature depth** — the investigation rests on ~6 primary mechanistic papers; additional GWAS fine-mapping and functional validation would strengthen causal attribution.

Proposed Follow-up Experiments / Actions

1. **Genotype-stratified clinical cohort study** — correlate rs7554511/p.Y333F carriage with IBD subtype, age at onset, behavior (B1/B2/B3), extraintestinal manifestations, and surgery rates.
2. **Fine-mapping and functional enhancer dissection** of the rs7554511 region (CRISPRi/MPRA) to confirm the causal regulatory element and transcription factors.
3. **Knock-in allelic-series mice** (p.Y333F and expression-lowering alleles) to model partial loss and test challenge-dependent colitis (DSS, TNF-α, enteric infection).
4. **Patient iPSC organoid barrier/immune co-cultures** to quantify permeability and macrophage clearance under risk vs protective genotypes, with ROCK/p38α/mTOR inhibitor rescue.
5. **Replicate and extend the infliximab pharmacogenomics** in larger multi-ancestry IBD cohorts; test whether INAVA genotype predicts biologic response.
6. **Microbiome × genotype interaction studies** to identify specific pathobionts that precipitate inflammation in INAVA-deficient hosts.
7. **Condensate-targeting proof-of-concept** — test whether modulating INAVA condensate dynamics shifts the barrier/inflammation balance in vivo.